COHOMOLOGY SHEAVES OF R LIM UNDER THE SUITABLE-SYSTEM CONDITIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Per-open triviality. Cohomology of Sheaves, Lemma cohomology-lemma-cohomology-over-U-trivial. H^q(U,K_n) = H^0(U,H^q(K_n)).
- Milnor sequence. Cohomology of Sheaves, Lemma cohomology-lemma-RGamma-commutes-with-Rlim. H^q(U, R lim K_n) = lim H^q(U,K_n) when R^1 lim vanishes.

Source reference. Original lemma 0BKT: sources/stacks-chow/tags.txt, source lines 14571–14571. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0D60: sources/stacks-chow/tags.txt, source lines 16505–16505. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let (K_n) be an inverse system of objects of $D(\mathcal{O}_X)$. Let $\mathcal{B}$ be a set of opens of X. Assume*

(1) *every open of X has a covering whose members are elements of $\mathcal{B}$,*
(2) *for all $U \in \mathcal{B}$ and all $q \in \mathbf{Z}$ we have*
   (a) *$H^p(U, H^q(K_n)) = 0$ for $p > 0$,*
   (b) *the inverse system $H^0(U, H^q(K_n))$ has vanishing $R^1 \lim$.*

Then $H^q(R \lim K_n) = \lim H^q(K_n)$ for $q \in \mathbf{Z}$.

Proof. Set $K = R \lim K_n$. Let $U \in \mathcal{B}$. By Lemma 0BKT and (2)(a) we have $H^q(U, K_n) = H^0(U, H^q(K_n))$. By Lemma 0D60 and (2)(b) we have $H^q(U, K) = \lim H^0(U, H^q(K_n))$. Thus $H^q(U, K)$ is the inverse limit the sections of the sheaves $H^q(K_n)$ over U. Since $\lim H^q(K_n)$ is a sheaf we find using assumption (1) that $H^q(K)$, which is the sheafification of the presheaf $U \mapsto H^q(U, K)$, is equal to $\lim H^q(K_n)$. This proves the lemma. □

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